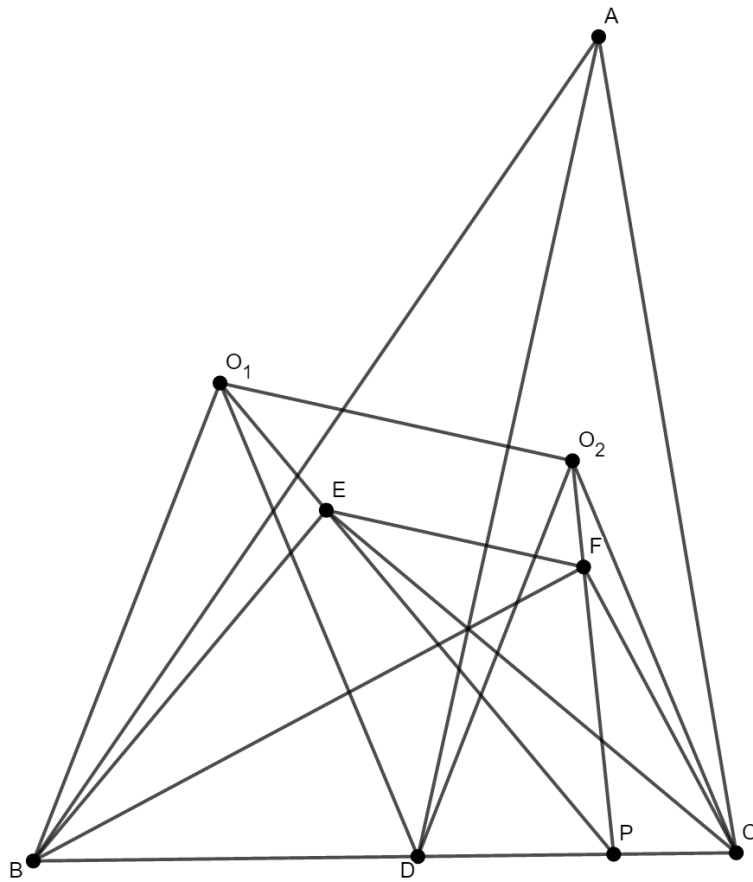


4.3.29



Obviously, O_1O_2 is the perpendicular bisector of AD . By simple angle chasing, $EF \perp AD$. So $O_1O_2 \parallel EF$.

$$\frac{d(O_1, BC)}{d(O_2, BC)} = \frac{AB}{AC} = \frac{d(E, BC)}{d(F, BC)}$$

so O_1E and O_2F meet on BC .